Joseph Li

Los Angeles, CA | (734) 934-2168 | lijoseph894@gmail.com | github.com/josephl894 | TN-1 Visa Eligible

Education

University of Southern California Sep 2024 - May 2026

MSE, Computer Science Engineering - Artificial Intelligence

- GPA: 3.50/4.00
- Relevant Coursework: Machine Learning, Natural Language Processing, Analysis of Algorithms, Database Systems

University of Michigan, Ann Arbor Sep 2020 - May 2024

BSE, Data Science Engineering

- GPA: 3.68/4.00

Professional Experience

Yoyo Markets Aug - Dec 2025

AI Engineer *New York, NY (Remote)*

- Led a 3-person team to replace manual research and editorial workflows with an AI pipeline that mines real-time social signals and generates 50+ prediction markets per week, a 5x throughput increase over the prior manual process
- Designed a multi-agent ELO ranking system evaluating 200+ candidate markets per cycle, improving market quality scores by 30% and eliminating manual curation from the ideation pipeline
- Built a custom Twitter scraper targeting 150+ high-signal KOL accounts using async Python, expanding the pipeline's real-time coverage beyond public trend aggregators and increasing signal volume by 3x
- Shipped a Next.js internal dashboard on Vercel used daily by the founding team to review, approve, and publish live markets, reducing end-to-end publishing time from 30 minutes to under 5

Pure Global Jun - Aug 2025

Machine Learning Engineering Intern *Jersey City, NJ (Remote)*

- Built a RAG-powered AI agent that classifies medical devices into FDA regulatory pathways (510(k), PMA, De Novo), cutting manual compliance research time by over 90%
- Engineered the end-to-end pipeline in n8n with retrieval-augmented generation over the FDA database API, indexing 150,000+ regulatory documents and achieving 88% classification accuracy on internal benchmarks

HIKE Investment Management Co. Jun - Aug 2024

Venture Intern *Beijing, China*

- Led due diligence on a Series D semiconductor startup with a team of five and authored a quarterly LP report on Nvidia's A100 supply chain and the US-China AI landscape, directly informing the fund's compute infrastructure thesis

Research & Projects

Privacy-Preserving NLP for Clinical Data | UMich SOCR Lab

- Developed a data obfuscation algorithm for unstructured clinical text using named entity recognition and differential privacy techniques, enabling safe use of 10,000+ patient records in published research
- Evaluated privacy-utility tradeoffs across 3 NLP pipelines, achieving 94% entity redaction accuracy while preserving downstream task performance within 3% of the unmasked baseline

Autonomous Research Agent (Personal Project)

- Built an agentic research tool in Python using LangGraph that decomposes a research question into sub-queries, searches arXiv and web sources in parallel, and synthesizes findings into a structured report with citations
- Implemented tool-calling orchestration, FAISS vector store for source deduplication, and streaming output; github.com/josephl894

Technical Skills

Languages: Python, C++, TypeScript, SQL
Frameworks & Libraries: PyTorch, LangChain, LangGraph, Next.js, FastAPI, Flask, pandas, NumPy, scikit-learn
Tools & Platforms: Git, Docker, AWS (S3, Lambda), PostgreSQL, FAISS, n8n, Vercel, CI/CD, pytest
Techniques: RAG, Multi-Agent Systems, NLP, Prompt Engineering, Fine-Tuning, Vector Databases